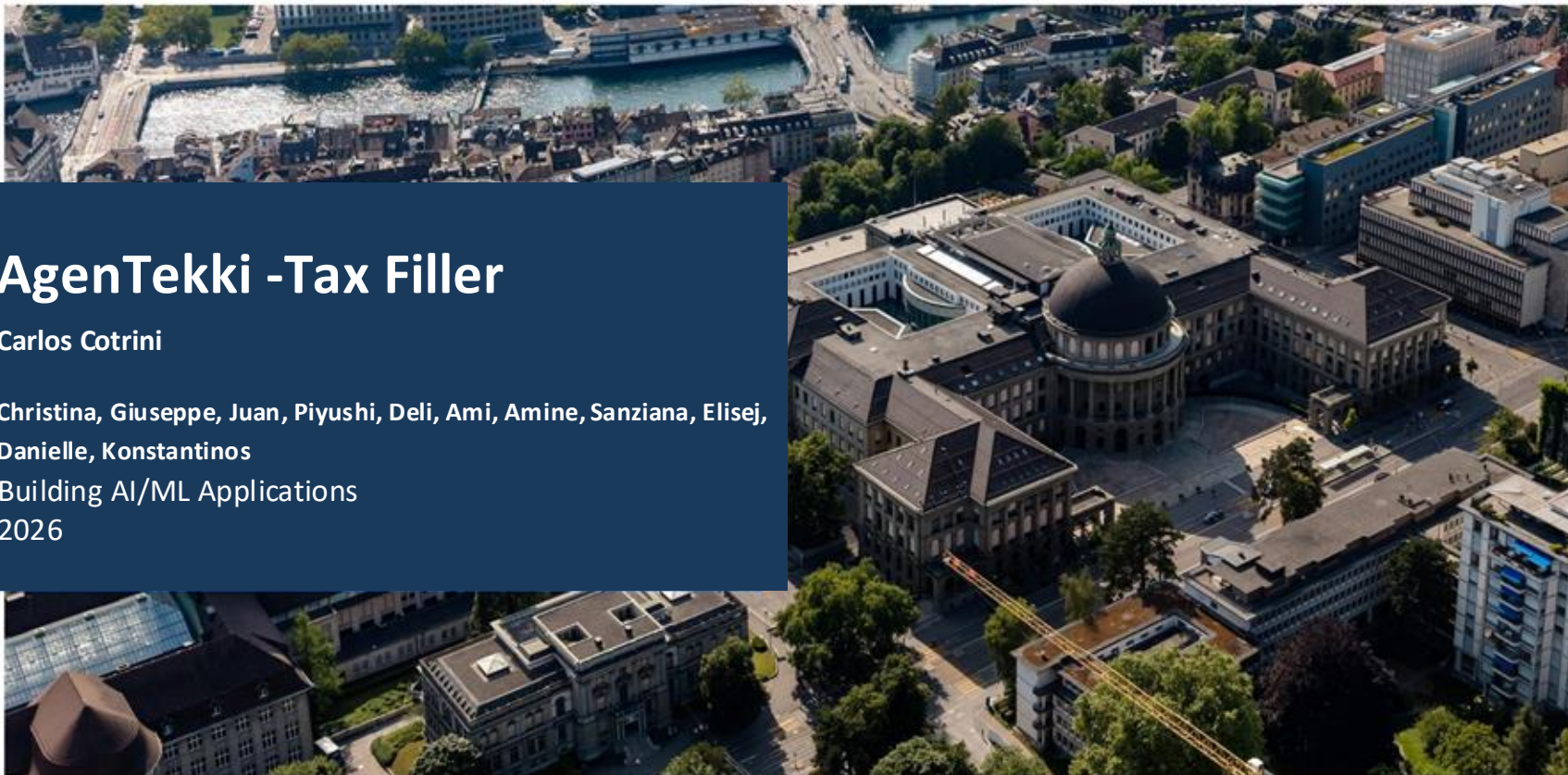


Agentekki -Tax Filler

Carlos Cotrini

Christina, Giuseppe, Juan, Piyushi, Deli, Ami, Amine, Sanziana, Elisej,
Danielle, Konstantinos
Building AI/ML Applications
2026



Meet the characters



The tax expert agent

An LLM that reads documents, looks up rules, reasons about the taxpayer's situation, and decides what goes in each field.

files taxes for →



Anna Meier

Single, easy



Bernasconis

Married couple, mortgage



Priya C.

Expat + freelance



Widmers

Retirement



Yuki T.

Divorce, expert

The 5 taxpayers (personas)

The goal

Your agent fills a simplified Zurich tax return accurately, autonomously, for each persona.

Personal

Income

Deductions

Wealth

Attachments

Review

1

Read documents

Lohnausweis, bank CSV, receipts, profile

2

Understand tax rules

Deduction limits, thresholds, formulas

3

Fill every field

Map the right value to the right field

4

Submit accurately

For all 5 personas, from easy to expert case

5 Taxpayers. 5 Levels of Complexity.



Anna Meier

Easy

**Software Engineer ·
Google Zurich**

One employer, one bank
account

Pillar 3a + ZVV tram pass

All data in the documents

*If your agent handles
Anna, you have the
fundamentals working.*



Bernasconis

Medium

**Marco (ABB) + Laura
(part-time physio)**

Joint filing, dual income

Childcare deduction cap

Prorate Laura's meal
allowance

*Property in Winterthur
with mortgage.*



Priya Chakraborty

Hard

**Data Scientist · Swiss Re
+ Freelance ML**

Mixed employment +
self-employment

USD investments → CHF
conversion

Capital gains are tax-free
in CH

*Expat from India, B
permit.*



Widmers

Very Hard

Thomas retired mid-year

8m salary + 4m AHV + 4m
BVG pension

CHF 250k capital
withdrawal

Capital withdrawal on
separate form!

*Critical: capital
withdrawal ≠ ordinary
income.*



Yuki Tanaka

Expert

**UX Designer · Spotify
Zurich**

Spousal support = taxable
income

Child support = NOT
income

Blurry KiTa receipt, no
property docs

*Recently divorced, one
child, Japanese citizen.*

The agentic AI framework

Every agentic system defines three things: the environment, the model, and the tools.

ENVIRONMENT

The Tax Portal

Also Includes: Personal documents
and Tax Guides!

External environment the Agent has to
navigate.

MODEL

The Large Language Model – LLM

Prompt (Text Input) → Text Output

The 'brain' that reasons
about the taxpayer's situation

TOOLS & HARNESS

Python Code

Code that connect the Model and the
Environment

The 'eyes, hands and feet' that let the
Model see and act on the environment

Environment: The Tax Portal

 **ZHprivateTax** · Declaration 2025 - No persona set

Overview / Tax Return

Form Navigation

Review / Submit

Back to Overview

Personal

Taxpayer Details

Children

Supported Persons

Representative

Gifts / Inheritances Received

Gifts / Advance Inheritances Given

Capital Benefits

Bank Details for Refunds

Income

Employment

Pensions & Insurance

Securities Income

Property Income

Other Income

Deductions

Commuting Costs

Other Professional Expenses

Taxpayer Details

Personal Details

First Name *

Last Name *

Date of Birth *
09/18/2026

Street *

Number *

Apartment / Suite

ZIP Code *

City *

Phone

Email

Marital Status *

Religion _

AHV Number *

Occupation *

Employer

Workplace

Environment Personal Documents

anna_meier/ Easy



profile.json

personal details



lohnusweis.txt

salary certificate



bank_statement.csv

transactions, interest, donations



pillar_3a.txt

Pillar 3a confirmation



zvv_receipt.txt

ZVV annual pass



private_notes.json

hints for the taxpayer NPC



ground_truth.json

correct answers for scoring

bernasconi/ Medium



profile.json

family details



lohnusweis_marco.txt

Marco's salary cert



lohnusweis_laura.txt

Laura's salary cert (60%)



bank_statement_joint.csv

joint account + mortgage



kita_receipt.txt

childcare costs (→ capped at 10,100)



physiouisse_membership.txt

professional expenses



property_assessment.txt

Eigenmietwert + Steuerwert



ground_truth.json

correct answers for scoring

Key Principles



Zero prior knowledge

The agent has no idea who the persona is until it reads the files.



Complexity grows

Richer folders mean harder reasoning
more files, more edge cases.



Scored automatically

submitted_return.json is compared
against ground_truth.json field by field.



Python scoring function

Tells you exactly what your agent got
right, missed, or filled incorrectly.



Your job

Build the system that reads and
reasons. Not memorise the documents.

Model: LLM or even Galileo?

Prompt (Text Input)

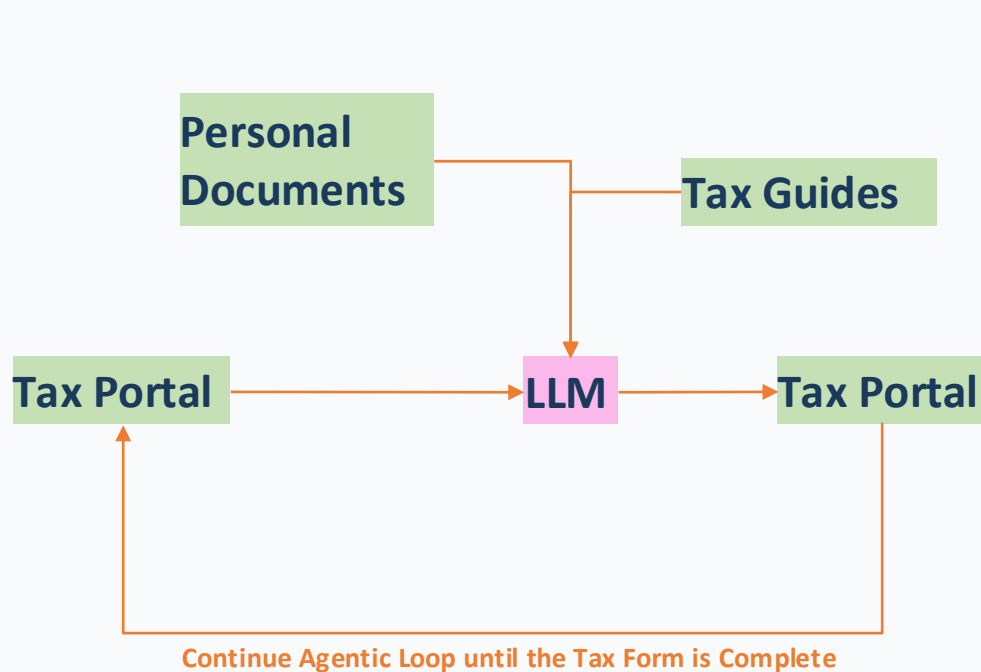


Text Output



Galileo is under House arrest and can only communicate via Letters (Text)

Putting it all together: Simplified Framework



Environment

Model

Orange Arrows Represent Tools

The Problem?

Requires large **Context window**.

Can be possible but is not efficient.

Framework: Simplified Tax Filing (In Terms of Galileo)

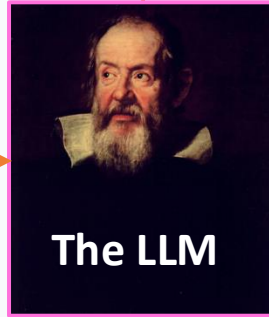
Personal Documents



Fig. 47. Brieftaube mit phot. Apparat.



Tax Guides



The LLM

Tax Portal

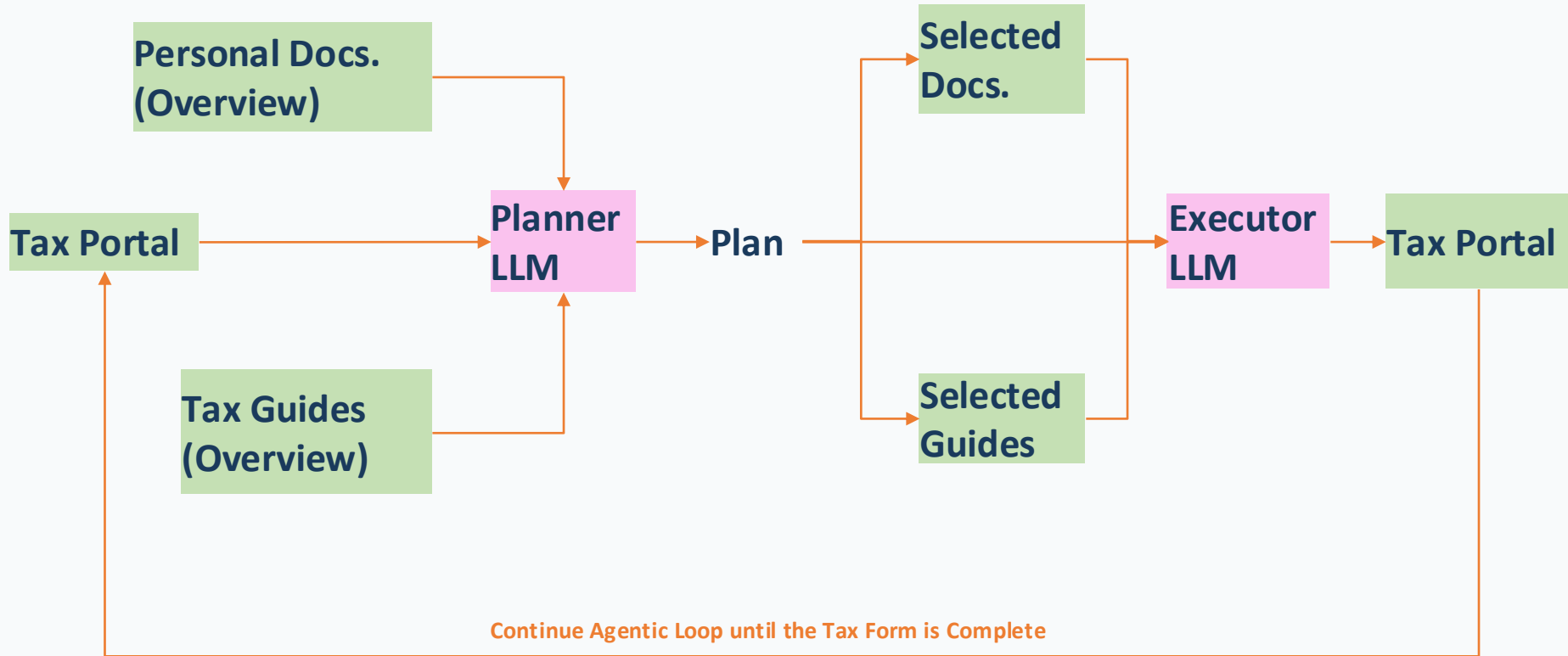
Tax Portal



The Problem:

Galileo is an Old man and gets confused by too much information at once.

Framework: (Advanced) Tax Agent



Tools for the Planner LLM:

- **scan_page()**

Reads the page as structured data: labels, field types, values, locators, buttons.

- **click_element(locator)**

Activates a portal control to navigate, add rows or submit.

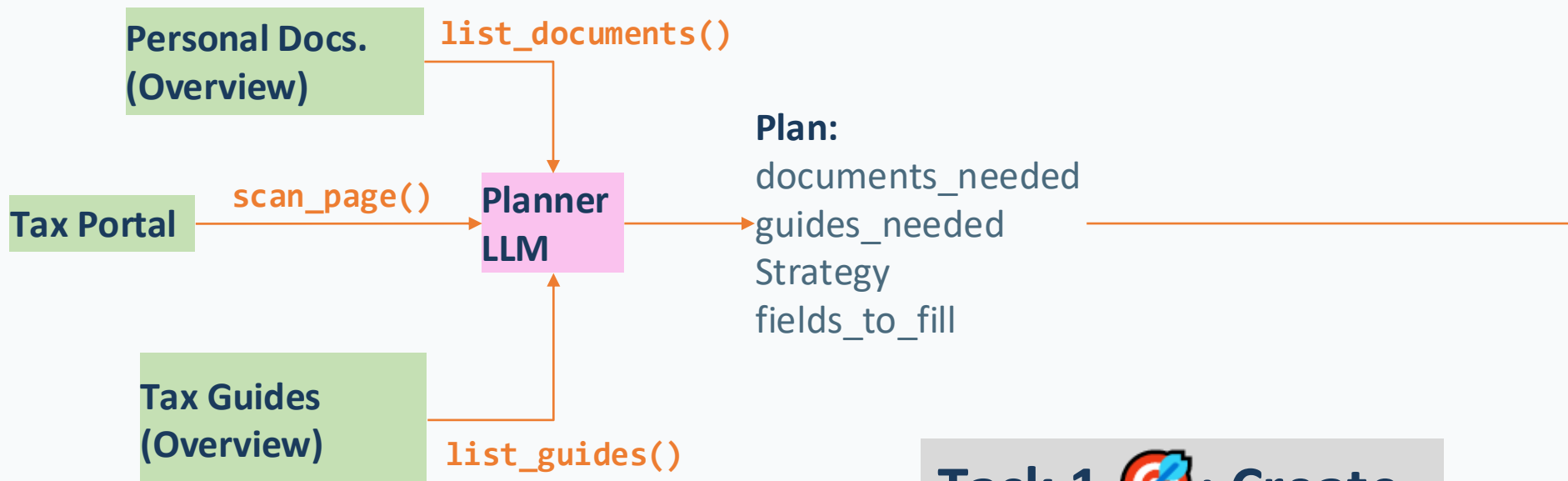
- **list_guides()**

Retrieves tax rules, thresholds and deduction guidance for a topic.

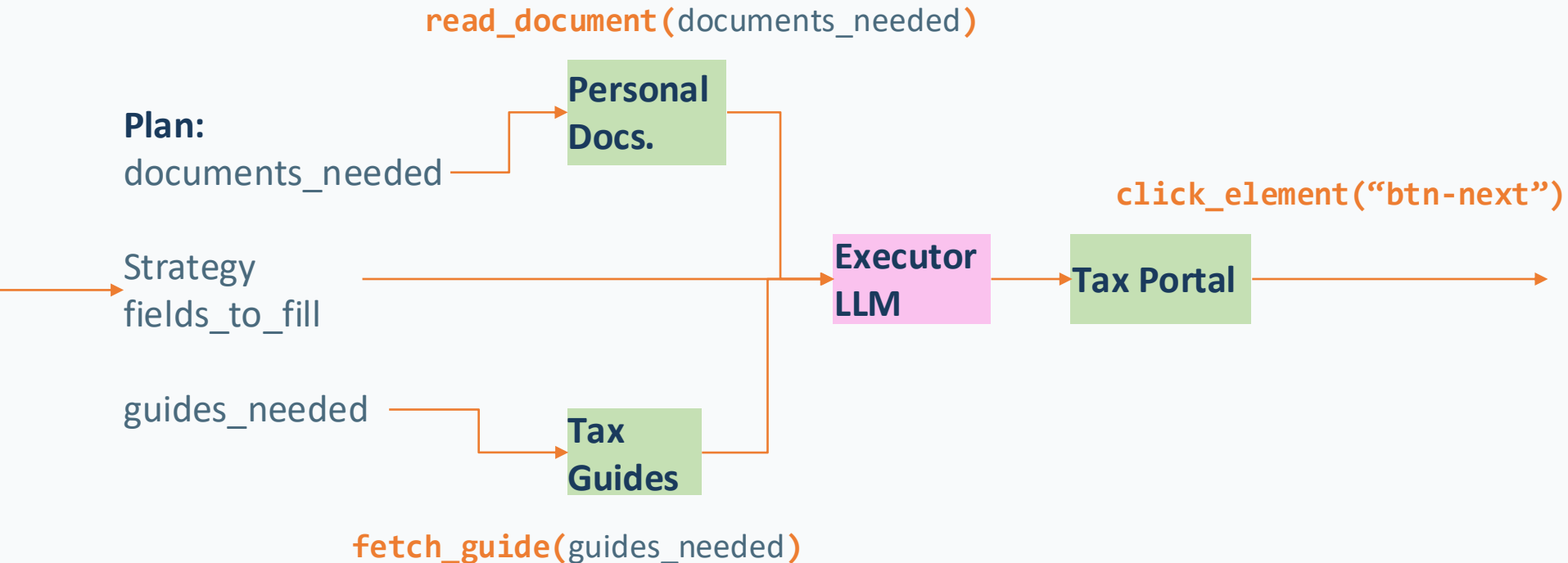
- **list_documents()**

Retrieves broad information about all the documents

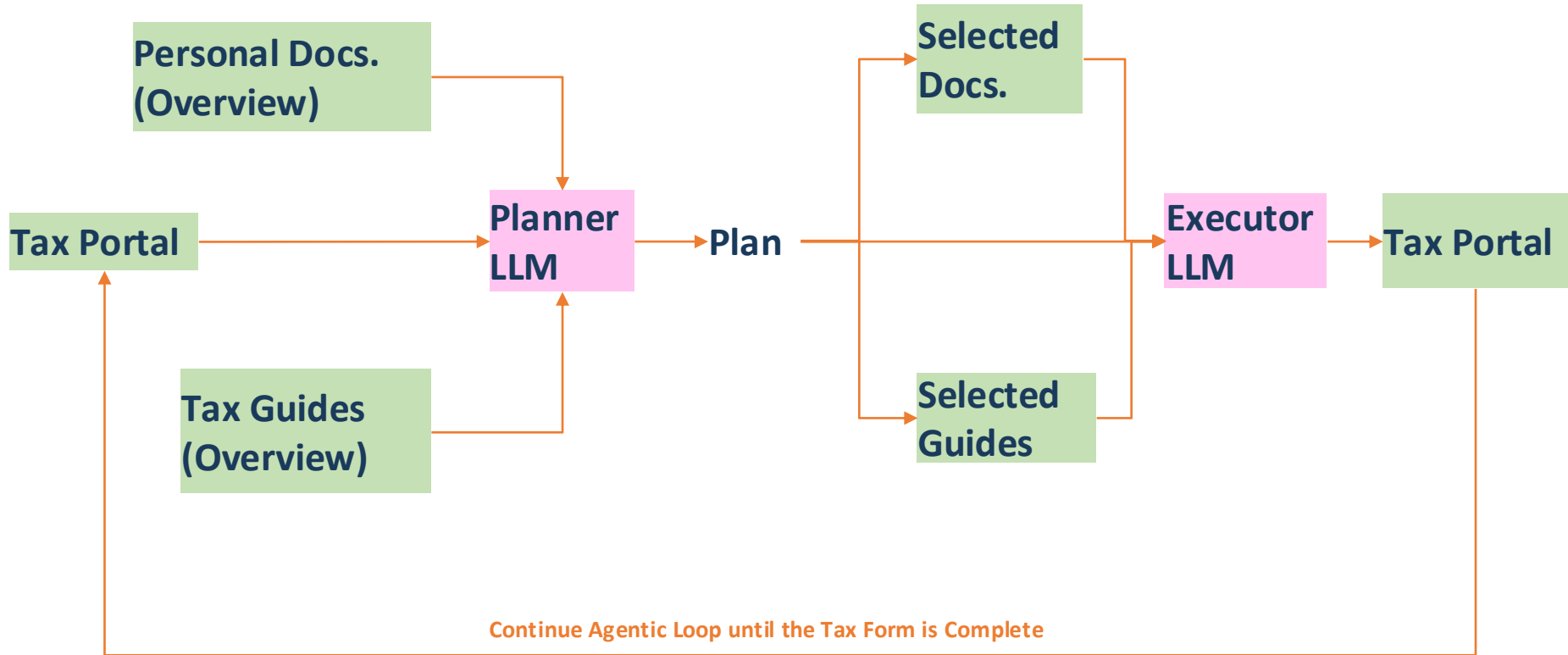
Tools in the Framework – Planner LLM



Tools in the Framework – Executor LLM



Framework: Tax Agent - Questions ?



Split up the Prompt: System Prompt and User Prompt



The System Prompt controls what kind of task the LLM performs.
Same model, different behavior.

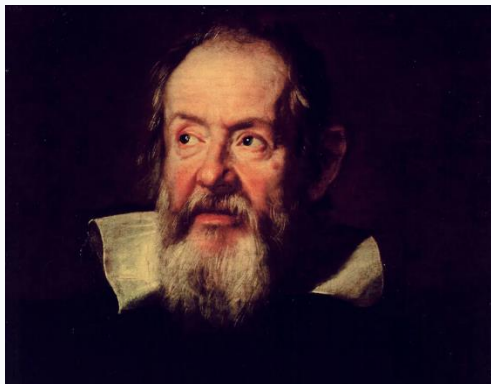
How is the Planner LLM created?

System Prompt:

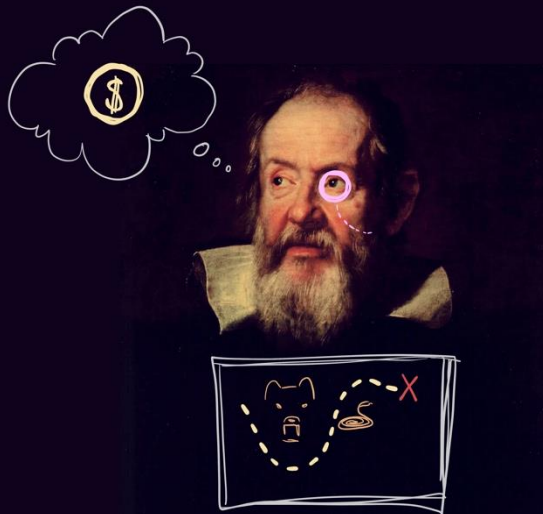
"You are an expert Swiss tax accountant AI.
You are **planning** how to fill one page of a Zurich Tax return."

System
Prompt

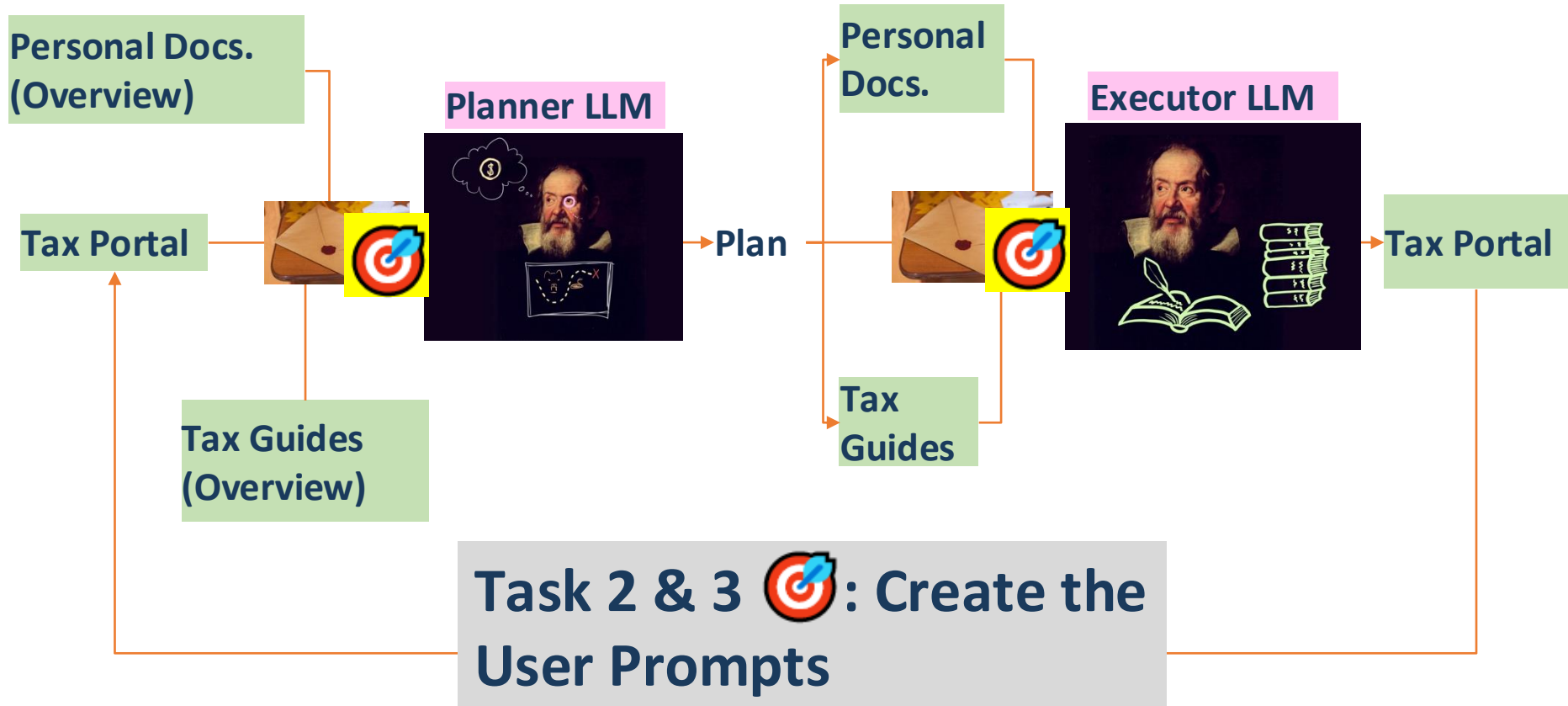
LLM



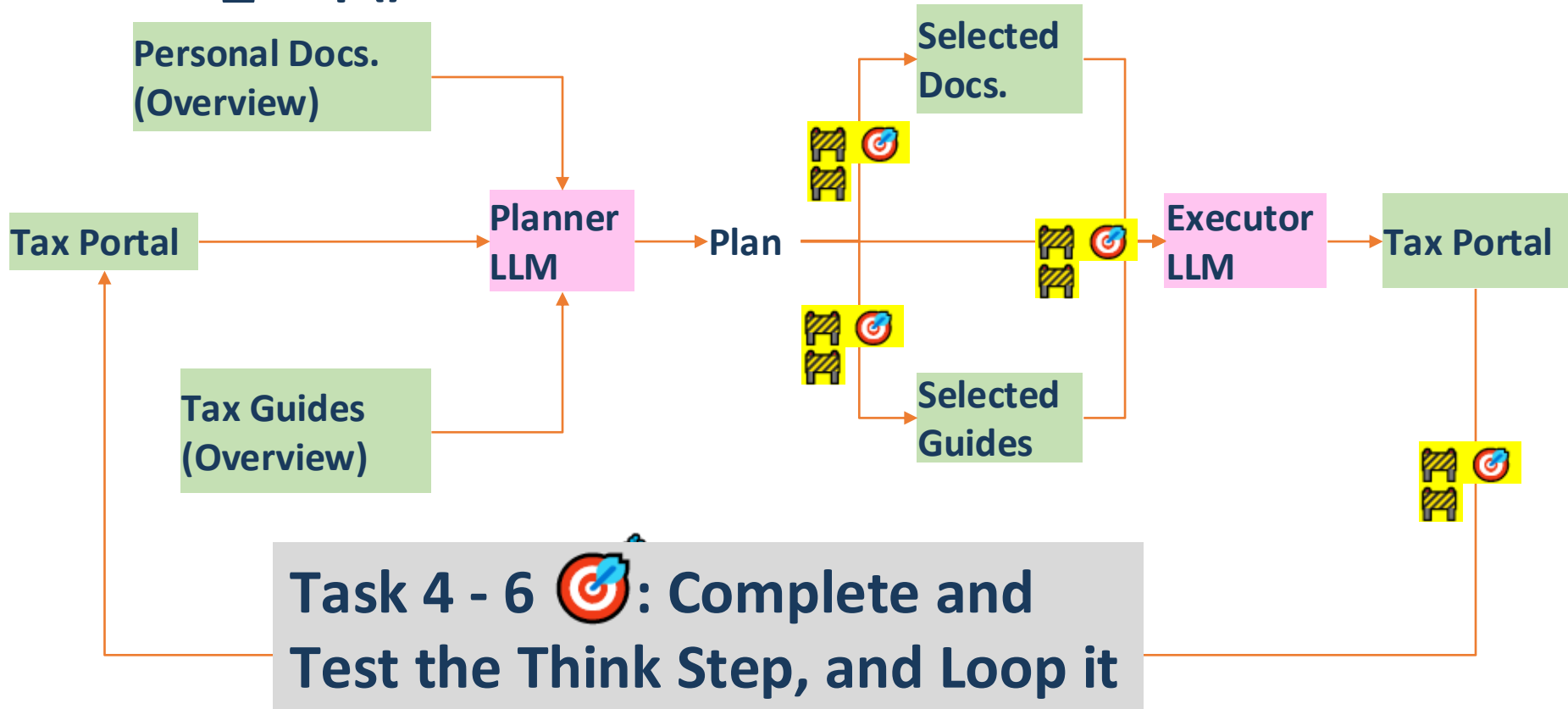
Planner LLM



Framework: Tax Agent (In Terms of Galileo)



We will call one iteration of this Loop the **think_step()**



Good Luck and we can't wait to see what you build!

Deadline: Sunday 25 October 2026, 23:59
(The Deadline for all projects is Sunday 25 October 2026, 23:59)